CURATED PATHWAY RECORD

Berberine Biosynthesis Pathway

Final target: ***Berberine*** | Current module: **M001** | Source database: **AA2_canonical_topology** | PMN ID: PWY-3901 |

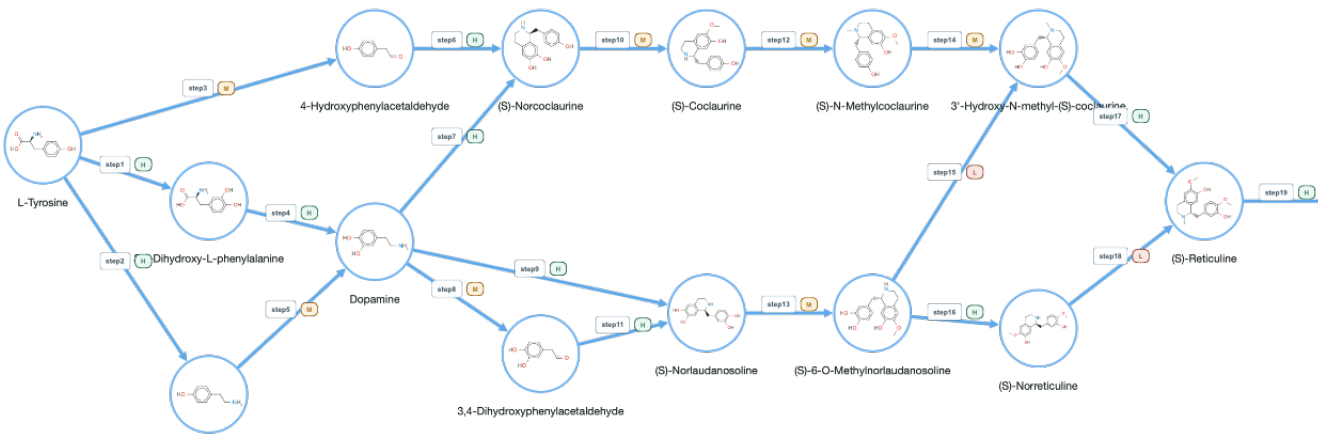
MODULES FOR BERBERINE M001 22 steps / 11 routes

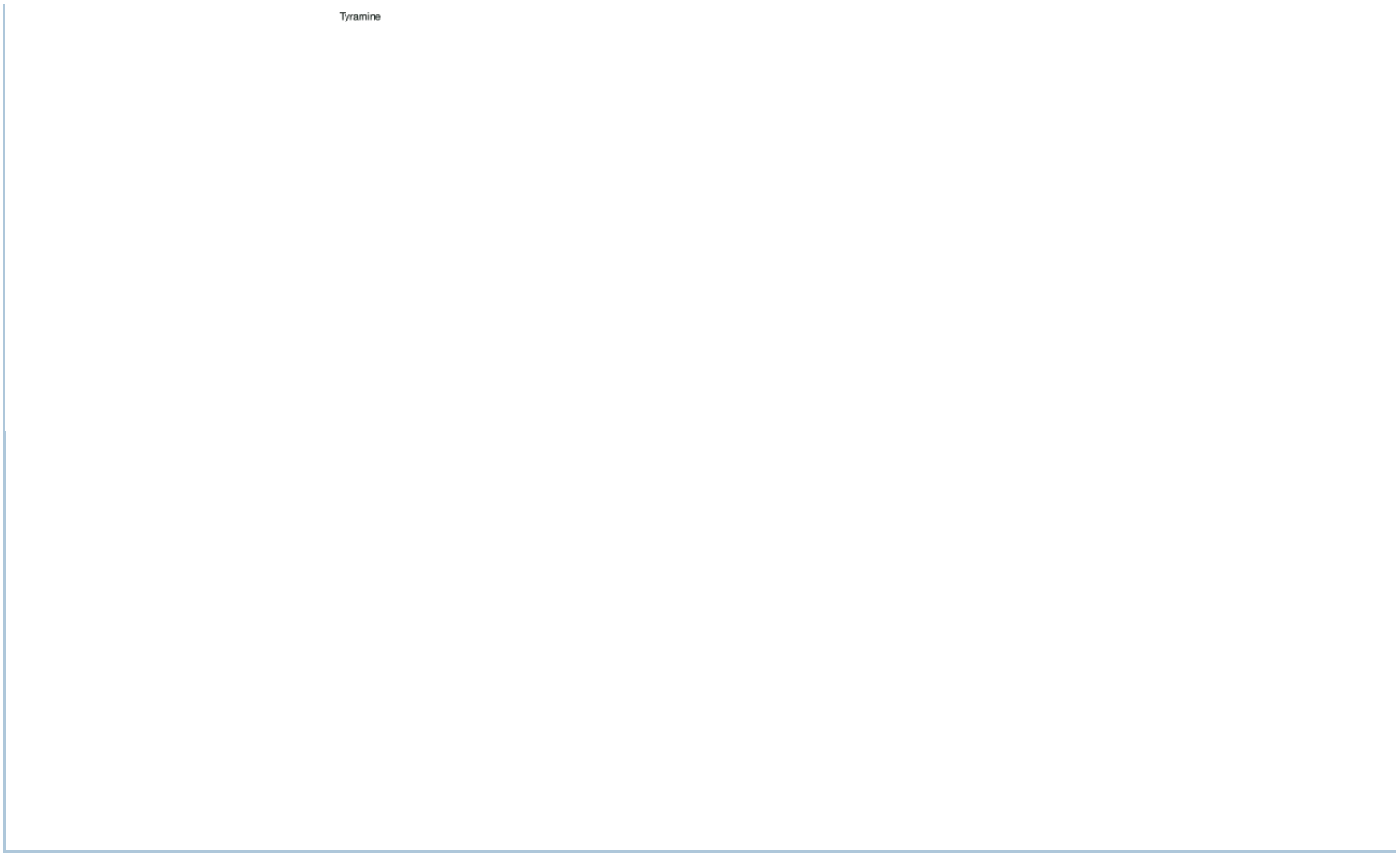
PATHWAY MAP

Berberine Biosynthesis Pathway Map

Compound nodes and reaction steps are shown for the current module. Selecting a route updates the map, evidence n
module.

ROUTE VIEW All routes





EVIDENCE MATRIX

Reaction steps and enzyme support

22 steps / **105** known enzymes / **5862** predicted enzymes / **5967** evidence records / **22** species-supported steps / m

SEARCH Step, compound, enzyme, EC SEED All seed levels SUPPORT All support

Step	Reaction	Substrate	Product	Enzyme
step1	C00082 -> C00355	L-Tyrosine	3,4-Dihydroxy-L-phenylalanine	tyrosinase +1 names
step2	C00082 -> C00483	L-Tyrosine	Tyramine	tyrosine decarbo +2 names
step3	C00082 -> C03765	L-Tyrosine	4-Hydroxyphenylacetaldehyde	-
step4	C00355 -> C03758	3,4-Dihydroxy-L-phenylalanine	Dopamine	tyrosine decarbo +1 names
step5	C00483 ->	Tyramine	Dopamine	tyrosinase

	C03758			
step6	C03765 -> C06160	4-Hydroxyphenylacetaldehyde	(S)-Norcoclaurine	(S)-norcoclaurine
step7	C03758 -> C06160	Dopamine	(S)-Norcoclaurine	(S)-norcoclaurine
step8	C03758 -> C04043	Dopamine	3,4-Dihydroxyphenylacetaldehyde	-
step9	C03758 -> C02916	Dopamine	(S)-Norlaudanosoline	(S)-norcoclaurine
step10	C06160 -> C06161	(S)-Norcoclaurine	(S)-Coclaurine	-
step11	C04043 -> C02916	3,4-Dihydroxyphenylacetaldehyde	(S)-Norlaudanosoline	(S)-norcoclaurine
step12	C06161 -> C05176	(S)-Coclaurine	(S)-N-Methylcoclaurine	-
step13	C02916 -> C06517	(S)-Norlaudanosoline	(S)-6-O-Methylnorlaudanosoline	-
step14	C05176 -> C05202	(S)-N-Methylcoclaurine	3'-Hydroxy-N-methyl-(S)-coclaurine	-
step15	C06517 -> C05202	(S)-6-O-Methylnorlaudanosoline	3'-Hydroxy-N-methyl-(S)-coclaurine	-
step16	C06517 -> C06520	(S)-6-O-Methylnorlaudanosoline	(S)-Norreticuline	3'-hydroxy-N-met coclaurine 4'-O-methyltransferas
step17	C05202 -> C02105	3'-Hydroxy-N-methyl-(S)-coclaurine	(S)-Reticuline	3'-hydroxy-N-met coclaurine 4'-O-methyltransferas
step18	C06520 -> C02105	(S)-Norreticuline	(S)-Reticuline	-
step19	C02105 -> C02106	(S)-Reticuline	(S)-Scoulerine	reticuline oxidas
step20	C02106 -> C04118	(S)-Scoulerine	Isocorypalmine	(S)-scoulerine 9- methyltransferas
step21	C04118 -> C03329	Isocorypalmine	(S)-Canadine	-

step22	C03329 -> C00757	(S)-Canadine	Berberine	tetrahydroprotob oxidase

POTENTIAL PRODUCER SPECIES

Route completeness

Complete producers are calculated per detected route and update with the selected route view.

44 complete species

ID	Route	Species	Chinese name	Completeness
1	Route 1	Cinnamomum kanehirae	沉水樟	**100%**
2	Route 1	Coptis chinensis	黄连	**100%**
3	Route 1	Macleaya cordata	博落回	**100%**
4	Route 1	Papaver somniferum	罂粟	**100%**
5	Route 2	Cinnamomum kanehirae	沉水樟	**100%**
6	Route 2	Coptis chinensis	黄连	**100%**
7	Route 2	Macleaya cordata	博落回	**100%**
8	Route 2	Papaver somniferum	罂粟	**100%**
9	Route 3	Cinnamomum kanehirae	沉水樟	**100%**
10	Route 3	Coptis chinensis	黄连	**100%**
11	Route 3	Macleaya cordata	博落回	**100%**
12	Route 3	Papaver somniferum	罂粟	**100%**
13	Route 4	Cinnamomum kanehirae	沉水樟	**100%**
14	Route 4	Coptis chinensis	黄连	**100%**
15	Route 4	Macleaya cordata	博落回	**100%**
16	Route 4	Papaver somniferum	罂粟	**100%**
17	Route 5	Cinnamomum kanehirae	沉水樟	**100%**

18	Route 5	Coptis chinensis	黄连	**100%**
19	Route 5	Macleaya cordata	博落回	**100%**
20	Route 5	Papaver somniferum	罂粟	**100%**

NGDC

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